

# Learnings: Innovating classic mobile games supported by data-driven insights with BigQuery

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

## ABOUT LEARNINGS



To continue creating content tailored to user preferences, Learnings leverages the data capabilities of BigQuery to run in-depth user analyses.

## About Learnings

Founded in 2016, Learnings is a Chinese game developer dedicated to innovating classic mobile games. It offers a wide range of game categories from coloring and puzzles to digital card games, and currently has more than 100 games on the market and around 70 million monthly active users worldwide. In 2021, two of Learnings' games were listed among the 30 most downloaded Chinese mobile games overseas for three consecutive months.

Industries: Technology

## Google Cloud results

- Enables meticulous A/B testing for more effective product development strategies with BigQuery
- Accurately identifies app flaws through user behavior analysis with BigQuery
- Reduces in-app content download time from seconds to milliseconds with Cloud CDN
- Supports smooth daily work coordination among 200 employees

Increase acquisition by mobile 50%

Location: China

CDN (<https://cloud.google.com/cdn/>), Cloud



## About WebEye

WebEye is one of the largest information distribution companies in China, covering a wide range of expertises from digital marketing to mobile game distribution. As a part of its services to help Chinese companies strive in markets abroad, WebEye also offers cloud infrastructure modernization solutions. In 2019, WebEye became a Google Cloud Premier Partner.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine  
(<https://cloud.google.com/compute/>)

Firebase (<https://firebase.google.com/>)

Google Workspace  
(<https://workspace.google.com/>)

across three  
office  
locations  
with Google  
Workspace

New mobile games enter the market all the time, but evergreen, classic genres, like coloring, puzzles, and elimination games continue to stay as players' all-time favorites. Statistics show that these classic mobile games have higher user retention rates (<https://www.blog.udonis.co/mobile-marketing/mobile-games/mobile-gaming-statistics>) on average compared to newer games.

On a mission to bring simple joys to everyone's life through games, Learnings (<https://learnings.com.cn/#/?lan=en-US>) was founded in China in 2016 and dedicated itself to innovating classic mobile games with more appealing content and features. As of August 2021, the company has more than 10 games on the market, including coloring, puzzle, elimination, and digital card games, with around 70 million monthly active users worldwide. In 2021, two of its games, Paint by Number and Block Puzzle saw more than 100 million downloads each and were listed (<https://inf.news/en/game/0282cf386e5f3d8f7ea08ba71d470677.html>) among the 30 most downloaded Chinese mobile games overseas for four consecutive months, in February to May.

“We believe that leveraging the power of data enables us to create the best possible game content for our users. Google Cloud provides all the data analytics tools we need to achieve that and seamlessly connects to the Google Analytics services we have been using.”

—John Li, co-founder &  
COO, Learnings

John Li, COO at Learnings, says that the company's success can be largely attributed to its data-driven mindset. When Learnings began developing its

games in 2017, it realized that it needed a powerful data analytics system to better understand its users and create products that perfectly cater to their preferences. To that end, Learnings chose to adopt Google Cloud (<https://cloud.google.com/>) because of its capabilities to support in-depth and complex data analysis, and the fact that the company had already been using several data-related Google services like Google Analytics (<https://marketingplatform.google.com/about/analytics/>), which made it easier to aggregate all its data and build a smooth data pipeline.

“We believe that leveraging the power of data enables us to create the best possible game content for our users,” explains Li. “Google Cloud provides all the data analytics tools we need to achieve that and seamlessly connects to the Google Analytics services we have been using.”

“BigQuery helps  
us understand  
our users at a  
much deeper  
level. By  
performing in-  
depth user  
behavior

analyses, we are  
capable of  
developing more  
appealing game  
features and  
doubling the  
number of our  
users in a few  
months.”

—John Li, co-founder &  
COO, Learnings

## **Using a powerful data analysis system to create attractive products and troubleshoot accurately**

With the technical support of WebEye (<https://www.webeye.com/english/home.html>), Learnings established a comprehensive data analysis and business intelligence system in six months on BigQuery (/bigquery), with Cloud SQL (/sql) and Cloud Storage (/storage) supporting data inquiry and storage. Li says that without all the ready-to-use tools provided by Google Cloud, the company would have spent more than a year to build its own data warehouse from scratch.

Respecting user privacy policies of the different countries and platforms, Learnings collects non-

sensitive user behavior data from its games, such as the time players download the games, make payments, and the length of time they spend on each content page, while ensuring that their personal identity remains anonymous. Thanks to the comprehensive data analytics capabilities of BigQuery, the company is able to understand its users' preferences better and subsequently create more attractive content. For example, the team found through funnel analysis that a large number of users of its coloring game app were more likely to return to the app when they completed filling colors in images of animals the previous day. With such interesting insight, the company knew that it had to enhance the quality and variety of the animal images in its game to retain more users.

The more thorough understanding of its users also allows Learnings to conduct more meticulous A/B testing, whereby two similar scenarios are presented with different variables, as it can classify its users with more specific criteria, such as their content preferences and frequency of use, and learn how different user groups receive a new feature in detail. This way, the team is able to create product development strategies that fully reflect players' needs.

"BigQuery helps us understand our users at a much deeper level. By performing in-depth user behavior analyses, we are capable of developing more appealing game features and doubling the number of our users in a few months," says Li.

On top of that, in-depth data analyses on BigQuery also help Learnings to troubleshoot more accurately and improve user experience. For example, when one of its game apps experienced a breakdown, the DevOps team used BigQuery to accurately identify the problem by looking at user behavior data on that app and swiftly solved the issue.



## **Seamless connecting databases to acquire more users with Firebase**

Before adopting BigQuery, Learnings had already been using Google Analytics, [Google AdMob](https://admob.google.com) (<https://admob.google.com>) and [Google Ads](https://ads.google.com) (<https://ads.google.com>) for marketing and advertising purposes. To enhance its marketing performance, the company employs [Firebase](https://firebase.google.com/) (<https://firebase.google.com/>), which links to all of its existing platforms to seamlessly import its advertising data to BigQuery for further analytics. By identifying the advertising ways that are most attractive to its potential users, Learnings has



successfully increased its user acquisition rate by more than 50%.

“Firebase can instantly connect and aggregate all our advertising data on Google services,” notes Li. “Without Firebase, it would have taken us at least half a year to develop a frontend software development kit to achieve the same result.”

## **Accelerating in-app content download speed globally with Cloud CDN**

Learnings also uses [Cloud CDN \(/cdn\)](#) to accelerate the download speed of its game content for users around the world. By offering edge caches that store content on local servers globally and thus allow faster connection, Cloud CDN has helped the company reduce the time required to download its in-app content from several seconds to milliseconds.

“As some of our users live in countries with slower internet connection, Cloud CDN is very important for us to ensure high connectivity for all our users. It has greatly enhanced the user experience of our apps and helped us expand in the global market.”

“There’s still a lot  
we can do with

the data we  
have. With the  
powerful  
analytics  
capabilities of  
Google Cloud,  
we're confident  
that we can fully  
unlock the value  
of data in the  
future and bring  
the most sought-  
after content  
and gaming  
experience to  
our players  
worldwide."

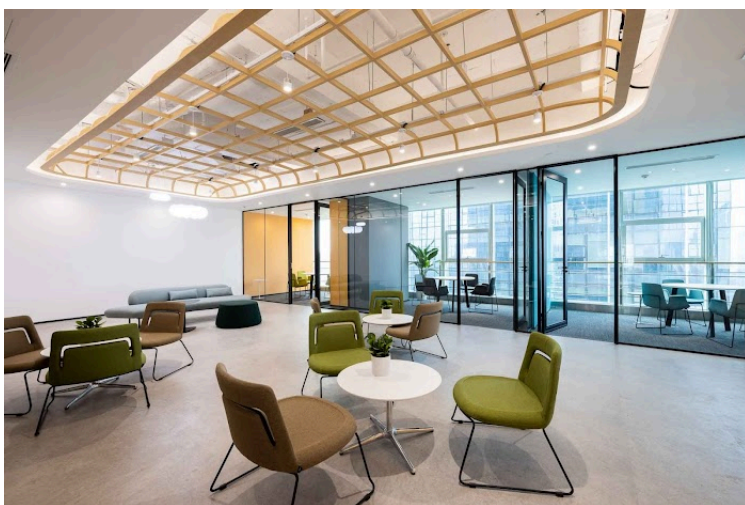
—John Li, co-founder &  
COO, Learnings

## **Seamless team collaboration across three office locations with Google Workspace**

Since its launch, Learnings has been relying on Google Workspace (<https://workspace.google.com/>) for internal communication and collaboration. As the company grows from 30 to more than 200

employees working in three different office locations in China, the collaborative editing feature of Google Docs (<https://workspace.google.com/products/docs/>) and Sheets (<https://workspace.google.com/products/sheets/>), the large storage volume of Drive (<https://workspace.google.com/products/drive/>), as well as the live video meetings accommodating more than 200 participants supported by Google Meet (<https://workspace.google.com/products/meet/>) have helped ensure the company's high work efficiency.

Furthermore, since all the Learnings' employees have a Gmail (<https://workspace.google.com/products/gmail/>) account, the company can effortlessly manage its authorization system on Google Cloud. "Our internal work system is fully integrated with Google Cloud, which enables us to easily streamline our workflow on the cloud platform," says Li.



**Further unlocking the value of data with artificial intelligence**

Moving forward, Learnings plans to take full advantage of BigQuery to perform a wider range of data analysis by understanding user profiles better, so it can generate preference predictions. As a part of its effort to deliver games that users enjoy the most, the company will also explore the artificial intelligence (AI) tools of Google Cloud to provide players with more personalized content recommendations.

Li says, “There’s still a lot we can do with the data we have. With the powerful analytics capabilities of Google Cloud, we’re confident that we can fully unlock the value of data in the future and bring the most sought-after content and gaming experience to our players worldwide.”